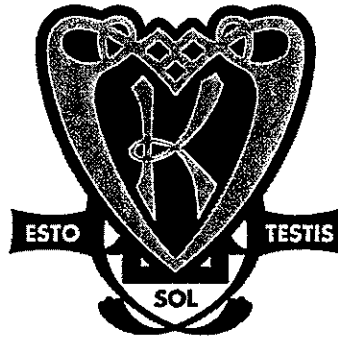


Student Number: _____

Class Teacher (*circle*): MC CG DL GP



YEAR 11 MATHEMATICS

Assessment Task 4

Preliminary Examination

September 2010

Instructions

- Time allowed: 2 hours plus 5 minutes reading time
- There are six questions, each worth 12 marks.
- The mark value of each part is indicated in [...] next to that part.
- Answer all questions on the writing paper provided. Start each question on a new page.
- Show all necessary working.
- Marks may not be awarded for careless or badly arranged work.

Question 1**Start a new page****[12 Marks]**

- (a) Find the value of m to three significant figures if: [2]

$$m = \sqrt{\frac{3.7 \times 8.4}{0.32}}$$

- (b) Factorise the expression $54 + 2x^3y^3$ [2]

- (c) Find the integers a and b such that $\frac{\sqrt{3}}{3+\sqrt{3}} = a + b\sqrt{3}$. [3]

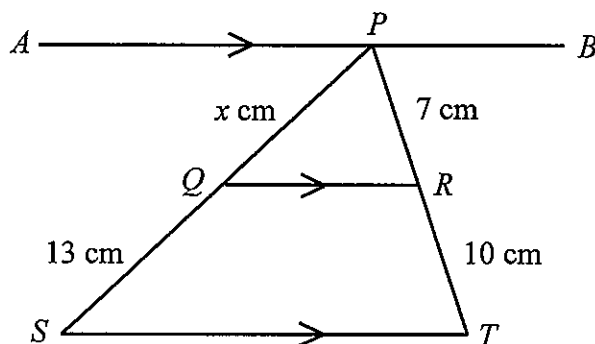
- (d) If $\tan \alpha = 4$ and $\cos \alpha < 0$, find the exact value of $\sin \alpha$ for $0 \leq \alpha \leq 360^\circ$. [3]

- (e) Solve the equation $\frac{p}{4} - \frac{p+3}{2} = 3$. [2]

Question 2**Start a new page****[12 Marks]**

- (a) Solve the inequation $x^2 + 5x - 6 > 0$. [3]

- (b) In the diagram below $AB \parallel QR \parallel ST$ and the lengths are as marked.



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Find the value of x to one decimal place, giving reasons for your answer. [2]

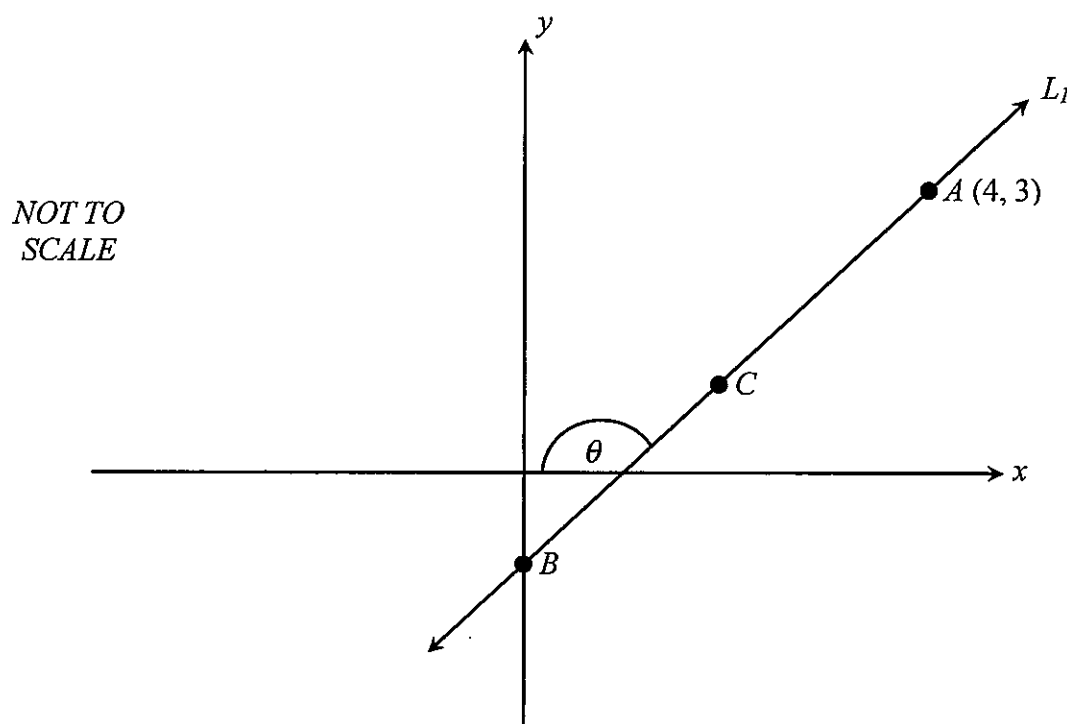
- (c) For the parabola $(x + 3)^2 = 12(y - 2)$
- (i) Find the coordinates of the vertex. [1]
 - (ii) Find the focal length. [1]
 - (iii) Find the coordinates of the focus. [1]
 - (iv) State the equation of the directrix. [1]
 - (v) Sketch the parabola labelling the vertex, focus and directrix. [1]
- (d) For what value(s) of p does $3x^2 - 4x + p = 0$ have equal roots? [2]

Question 3**Start a new page****[12 Marks]**

- (a) Solve $|2x - 4| < 6$. [2]
- (b) Differentiate the following:
- (i) $5x^4 + 2$ [1]
- (ii) $y = (2x - 5)^7$ [2]
- (iii) $f(x) = \frac{x+3}{2x-4}$ [2]
- (c) Simplify $\frac{\tan A \sec A}{1 + \tan^2 A}$. [2]
- (d) (i) On the same set of axes, sketch graphs of $x^2 + y^2 = 4$ and $x + y = 1$. Note: It is not necessary to find the point(s) of intersection of both graphs. [2]
- (ii) On your graph from (i) above, shade the region where the pair of inequalities $x^2 + y^2 \geq 4$ and $x + y \leq 1$ hold simultaneously. [1]

Question 4**Start a new page****[12 Marks]**

- (a) Points $A(4, 3)$, B and C lie on line L_1 as illustrated in the diagram below.



- (i) The equation of line L_1 is $x - y - 1 = 0$. Find the coordinates of B . [1]
- (ii) Find the size of θ , the obtuse angle between the x -axis and L_1 . [2]
- (iii) Find the coordinates of point C , the mid-point of AB . [2]
- (iv) Find the length of AB . [1]
- (v) Find the equation of the circle of which AB is a diameter. [2]
- (b) (i) On the same set of axes, sketch the graphs of $y = \cos x$ and $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$, showing all important features. [3]
- (ii) Hence or otherwise, determine the number of solutions to the equation $\tan x - \cos x = 0$ in the domain $0^\circ \leq x \leq 360^\circ$. [1]

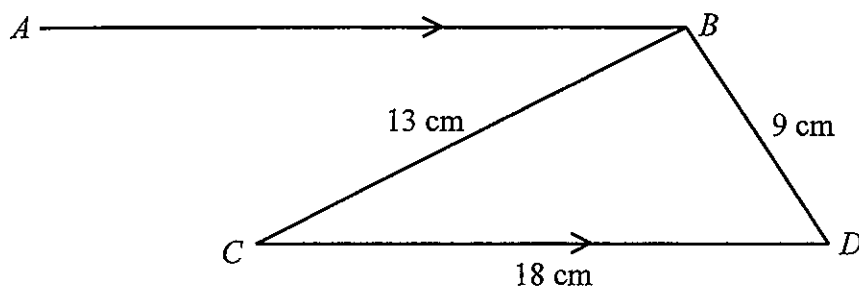
Question 5**Start a new page****[12 Marks]**

(a) Given the function $f(x) = x(x+3)^2$,

(i) Find $f'(x)$. [2]

(ii) Find the value(s) of x such that $f'(x) = 0$. [2]

(b) In the diagram below, AB is parallel to CD and the lengths of BC , BD and CD are as shown.



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(i) Find the size of $\angle BDC$ to the nearest degree. [2]

(ii) Hence find the size of $\angle ABD$, giving reasons for your answer. [1]

(c) Find $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$. [2]

(d) Find the shortest distance between the two parallel lines $l_1: x - 3y + 6 = 0$ and $l_2: x - 3y - 2 = 0$. [3]

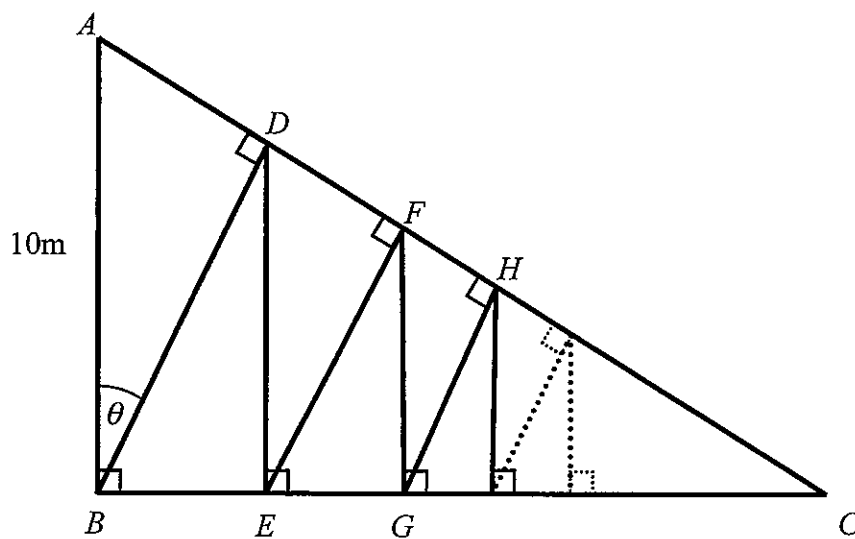
Question 6**Start a new page****[12 Marks]**

- (a) Find the values of a , b and c if: [3]

$$x^2 + x + 1 \equiv a(x-1)^2 + b(x-1) + c$$

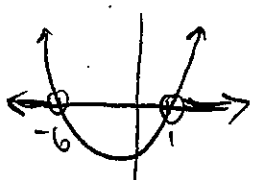
- (b) If the two roots of the equation $2x^2 - 5(p-1)x + 12 = 0$ are consecutive integers, find the value(s) of p . [3]

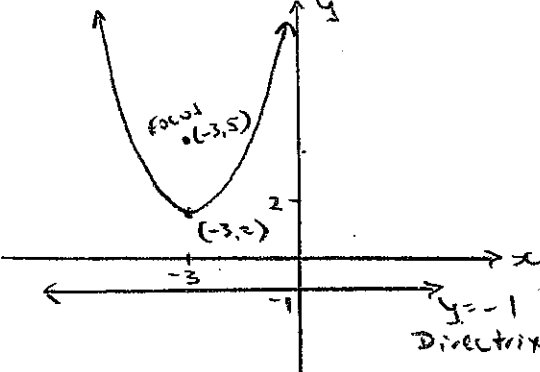
- (c) The diagram below shows a roof truss, comprising a number of right-angled triangles. $\angle ABD = \theta$ and the length of AB is 10m.



- (i) Prove that $\triangle ABD \parallel \triangle BDE$. [2]
- (ii) Show that $BD = 10 \cos \theta$. [1]
- (iii) Find an expression for the length of EF in terms of $\cos \theta$. [2]
- (iv) If $\theta = 30^\circ$, find the exact length of EF . [1]

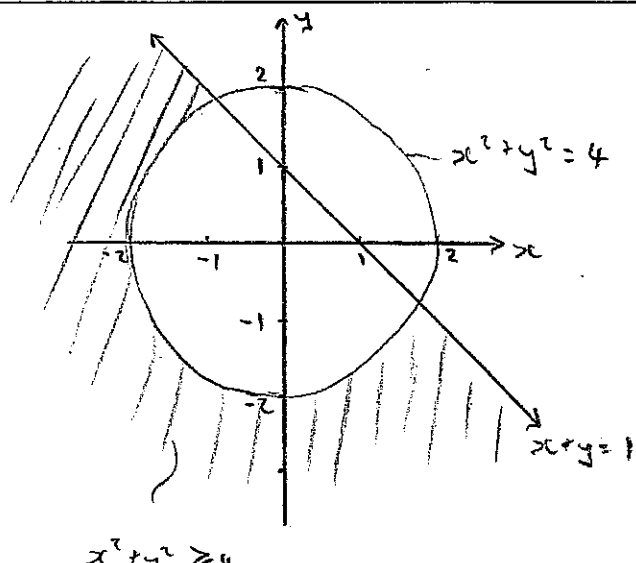
End of Assessment Task

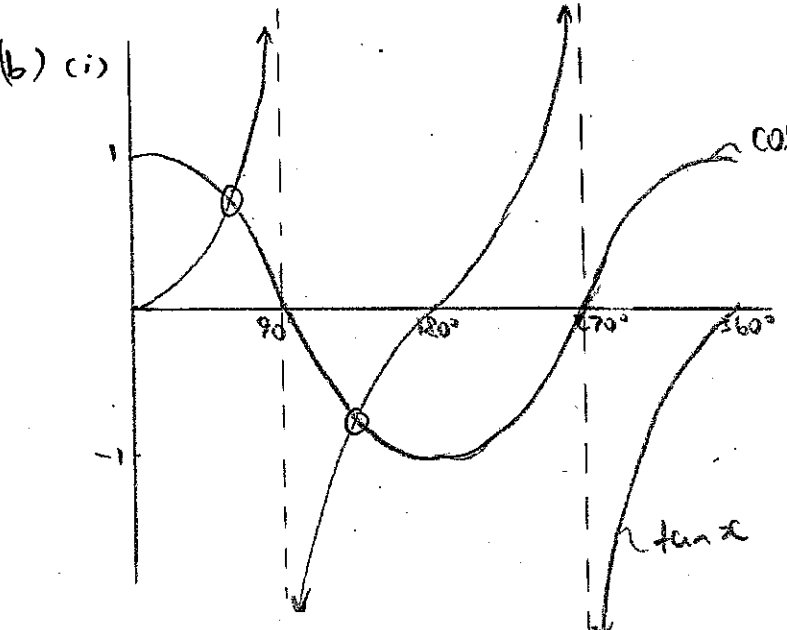
Qn	Solutions	Marks	Comments+Criteria
1	(e) $\frac{p}{4} - \frac{p+3}{2} = 3$ $p - 2(p+3) = 3 \times 4$ $p - 2p - 6 = 12$ or $-p - 6 = 12$ $-p = 18$ $p = -18$	1 1	for denominator correct answer
2	(a) $x^2 + 5x - 6 > 0$ $(x+6)(x-1) > 0$  $\therefore x < -6$ or $x > 1$ (b) $\frac{x}{13} = \frac{7}{10}$ (Ratio of intercepts on one transversal equals the ratio of intercepts on any other transversal) $10x = 13 \times 7$ $x = \frac{13 \times 7}{10}$ $= 9.1$ (c) $(x+3)^2 = 12(y-2)$ (i) Vertex: $(-3, 2)$ (ii) Focal Length: $4a = 12$ $\therefore a = 3$ Focal Length = 3	1 1 1 1 1 1	factors for each soln 1 for reason 1 off for any error correct answer Correct answer only Correct answer

Qn	Solutions	Marks	Comments+Criteria
2(c)	(iii) Parabola is concave up: $\therefore$ Focus = $(-3, 2+3)$ $= (-3, 5)$ (iv) Directrix: $y = 2-3$ $y = -1$ (v)  (d) $3x^2 - 4x + p = 0$ Equal roots $\Delta = 0$ $\Delta = b^2 - 4ac$ $0 = (-4)^2 - 4(3)(p)$ $0 = 16 - 12p$ $16 - 12p = 0$ $12p = 16$ $p = 16/12$ $= 4/3$	1 1 1 1 1 1	for answer for answer - must be eqn. for sketch set $\Delta = 0$ answer
3	(a) $2x - 4 < 6$ Either $2x - 4 < 6$ or $2x - 4 > -6$ $2x < 10$ $2x > -2$ $x < 5$ $x > -1$ $\therefore -1 < x < 5$	1	for each case -1 not continuous domain

domain

Qn	Solutions	Marks	Comments+Criteria
3	(b) (i) $\frac{d}{dx} 5x^4 + 2$ $= 20x^3$ (ii) $y = (2x-5)^7$ $\frac{dy}{dx} = 7(2x-5)^6 \cdot 2$ $= 14(2x-5)^6$ (iii) $f(x) = \frac{x+3}{2x-4}$ Quotient rule $u = x+3 \quad v = 2x-4$ $u' = 1 \quad v' = 2$ $f'(x) = \frac{vu' - uv'}{v^2}$ $= \frac{(2x-4) - 2(x+3)}{(2x-4)^2}$ $= \frac{2x-4-2x-6}{(2x-4)^2}$ $= \frac{-10}{(2x-4)^2}$	1	Correct answer
	(ii) $y = (2x-5)^7$ $\frac{dy}{dx} = 7(2x-5)^6 \cdot 2$ $= 14(2x-5)^6$	1	power down, reduce power
	(iii) $f(x) = \frac{x+3}{2x-4}$ Quotient rule $u = x+3 \quad v = 2x-4$ $u' = 1 \quad v' = 2$	1	Differentiate ()
	$f'(x) = \frac{vu' - uv'}{v^2}$ $= \frac{(2x-4) - 2(x+3)}{(2x-4)^2}$ $= \frac{2x-4-2x-6}{(2x-4)^2}$ $= \frac{-10}{(2x-4)^2}$	1	for set up
	(c) $\frac{\tan A \sec A}{1 + \tan^2 A} = \frac{\frac{\sin A}{\cos A} \times \frac{1}{\cos A}}{\sec^2 A}$	1	for substitution - sec ² A key
or	$\frac{\tan A \sec A}{1 + \tan^2 A}$ $= \frac{\tan A \sec A}{\sec^2 A}$ $= \frac{\tan A}{\sec A}$ $= \frac{\frac{\sin A}{\cos A}}{\frac{1}{\cos A}} = \frac{\sin A}{\cos A} \times \frac{\cos A}{1} = \sin A$	1	answer

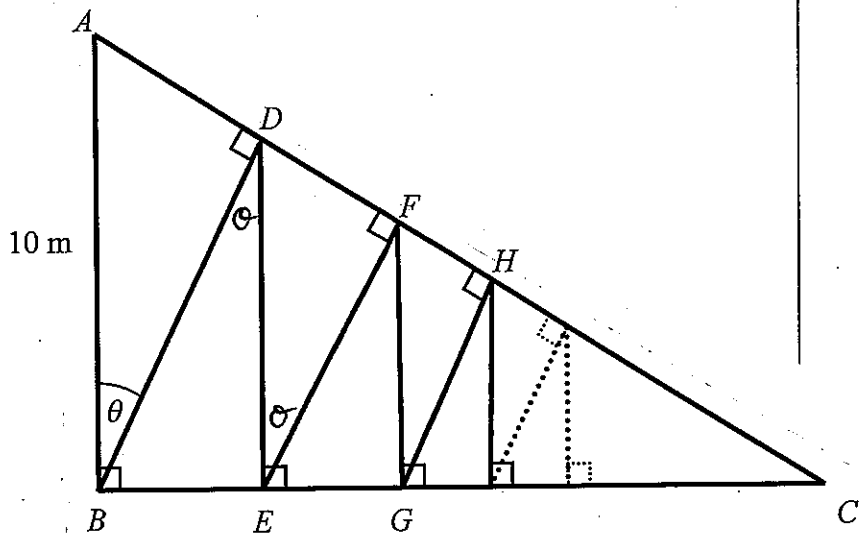
Qn	Solutions	Marks	Comments+Criteria
3	(d)  $x^2 + y^2 = 4$ $x + y = 1$ $x^2 + y^2 > 4$ and $x + y < 1$	1 1	for each graph Correct region
4	(a) (i) $x - y - 1 = 0$ $B - x = 0$ $\therefore 0 - y - 1 = 0$ $y = -1$ $\therefore B(0, -1)$ (ii) $x - y - 1 = 0$ i.e. $y = x - 1$ $\therefore m = 1$ $m = \tan \alpha$ (acute angle) $1 = \tan \alpha$ $\therefore \alpha = 45^\circ$ $\theta = 180 - 45$ $= 135^\circ$ (iii) $C \left(\frac{0+4}{2}, \frac{-1+3}{2} \right)$ $= (2, 1)$ (iv) $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $= \sqrt{(4 - 0)^2 + (3 - (-1))^2}$ $= \sqrt{4^2 + 4^2}$ $= \sqrt{32}$ $= 4\sqrt{2}$ units	1 1 1 1	Answer - must be coordinates for gradient for answer (recognise $180 - \theta$) x y } values at substitution point

Qn	Solutions	Marks	Comments+Criteria
4	(a) (v) C is $C(2, 1)$ $AB = 4\sqrt{2}$ units $= \text{diameter}$ $\therefore \text{Radius} = 2\sqrt{2}$ units Eg- of circle: $(x-2)^2 + (y-1)^2 = (2\sqrt{2})^2$ $(x-2)^2 + (y-1)^2 = 8$ (b) (i)  (ii) Solns to $\tan x - \cos x = 0$ i.e. $\tan x = \cos x$ $\therefore$ 2 solutions (intersection of graphs)	1 1 1 1 1	for centre for radius for shape $\cos x$ for shape $\tan x$ for axes values including asymptotes as per graph intersection
5	(a) $f(x) = x(x+3)^2$ (ii) $= x(x^2 + 6x + 9)$ $= x^3 + 6x^2 + 9x$ $f'(x) = 3x^2 + 12x + 9$ Product rule $f'(x) = uv' + uv'$ $u = x \quad v = (x+3)^2$ $u' = 1 \quad v' = 2(x+3)$ $f'(x) = (x+3)^2 + 2x(x+3)$ $= x^2 + 6x + 9 + 2x^2 + 6x$ $= 3x^2 + 12x + 9$ 2 differential -1 for error		1 for set up 1 for substitution

Qn	Solutions	Marks	Comments+Criteria
5(a)	(ii) $f'(x) = 0$ i.e. $3x^2 + 12x + 9 = 0$ $3(x^2 + 4x + 3) = 0$ $x^2 + 4x + 3 = 0$ $(x + 3)(x + 1) = 0$ $x = -3 \text{ or } -1$	1 1	factoring (progress) correct answer
(b)	(i) By Cosine rule Let $\angle BDC = \theta$ $\cos \theta = \frac{18^2 + 9^2 - 13^2}{2 \times 18 \times 9}$ $= \frac{59}{81}$ $\theta = \cos^{-1}\left(\frac{59}{81}\right)$ $= 43.2^\circ$ $= 43^\circ \text{ (nearest degree)}$ (ii) $\angle ASD = 180 - 43$ $= 137^\circ \text{ (co-interior angles are supplementary on parallel lines)}$	1 1	sub into cosine rule answer 1 for angle no penalty, if no reason or incorrect reason
(c)	$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ $= \lim_{x \rightarrow 2} \frac{(x/2)(x+2)}{(x/2)}$ $= \lim_{x \rightarrow 2} (x + 2)$ $= 2 + 2$ $= 4$	1 1	factoring answer

Qn	Solutions	Marks	Comments+Criteria																		
5	(d) $l_1: x - 3y + 6 = 0$ $l_2: x - 3y - 2 = 0$ Find a point on l_1, let $x = 0$ $0 - 3y + 6 = 0$ $3y = 6$ $y = 2$ $\therefore (0, 2)$ lies on l_1 $d = \left \frac{ax + by + c}{\sqrt{a^2 + b^2}} \right$ $= \left \frac{1(0) + (-3)(2) + (-2)}{\sqrt{(1)^2 + (-3)^2}} \right$ $= \left \frac{0 - 6 - 2}{\sqrt{10}} \right$ $= \left \frac{-8}{\sqrt{10}} \right$ $= \frac{8}{\sqrt{10}} \text{ units}$ $= \frac{8\sqrt{10}}{10}$ $= \frac{4\sqrt{10}}{5} \text{ units}$	1 1 1	for any correct point on one line Substitution answer																		
6	(a) $x^2 + x + 1 \equiv a(x-1)^2 + b(x-1) + c$ RHS $= a(x^2 - 2x + 1) + b(x-1) + c$ $= ax^2 - 2ax + a + bx - b + c$ $= ax^2 - (2a - b)x + (a - b + c)$ Equating coefficients <table><tr><td>$a = 1$</td><td rowspan="4"> </td><td>$a - b + c = 1$</td></tr><tr><td>$-(2a - b) = 1$</td><td>$1 - 3 + c = 1$</td></tr><tr><td>$-2a + b = 1$</td><td>$c = 3$</td></tr><tr><td>$-2(1) + b = 1$</td><td></td></tr><tr><td>$b = 3$</td><td></td><td>$\therefore a = 1$</td></tr><tr><td></td><td></td><td>$b = 3$</td></tr><tr><td></td><td></td><td>$c = 3$</td></tr></table>	$a = 1$		$a - b + c = 1$	$-(2a - b) = 1$	$1 - 3 + c = 1$	$-2a + b = 1$	$c = 3$	$-2(1) + b = 1$		$b = 3$		$\therefore a = 1$			$b = 3$			$c = 3$	1	for a, b, c Award marks for proper method
$a = 1$		$a - b + c = 1$																			
$-(2a - b) = 1$		$1 - 3 + c = 1$																			
$-2a + b = 1$		$c = 3$																			
$-2(1) + b = 1$																					
$b = 3$		$\therefore a = 1$																			
		$b = 3$																			
		$c = 3$																			

Qn	Solutions	Marks	Comments+Criteria
	Qr $x^2 + x + 1 \equiv a(x-1)^2 + b(x-1) + c$ Sub $x = 1$ $(1)^2 + (1) + 1 \equiv a(1-1)^2 + b(1-1) + c$ $3 \equiv c$ Equate coefficients of x^2 $x^2 = ax^2$ $\therefore a = 1$ Sub $x = 0$ $0 + 0 + 1 \equiv 1(0-1)^2 + b(0-1) + 3$ $1 \equiv 1 - b + 3$ $b = 3$ $\therefore a = 1, b = 3, c = 3.$ (b) $2x^2 - 5(p-1)x + 12 = 0$ Two roots are consecutive integers Let the roots be $\alpha, \alpha+1$ $\therefore \alpha(\alpha+1) = \frac{c}{a}$ $\alpha^2 + \alpha = \frac{12}{2}$ $\alpha^2 + \alpha - 6 = 0$ $(\alpha+3)(\alpha-2) = 0$ $\therefore \alpha = 2 \text{ or } -3$ Sum of roots: $\alpha + \alpha + 1 = \frac{-b}{a}$ $2\alpha + 1 = \frac{-[-5(p-1)]}{2}$ $2\alpha + 1 = \frac{5p-5}{2}$ $4\alpha + 2 = 5p - 5$ $5p = 4\alpha + 7$ $p = \frac{4\alpha + 7}{5}$ When $\alpha = 2$ or $\alpha = -3$ $p = \frac{4(2) + 7}{5} = \frac{15}{5} = 3$ $p = \frac{4(-3) + 7}{5} = \frac{-12 + 7}{5} = \frac{-5}{5} = -1$ $\therefore p = 3 \text{ or } -1$	1 1 1 or 1 for progress 1 for answer	Award marks for progress/method for a, b, c for $\alpha, \alpha+1$ factored quadratic

Qn	Solutions	Marks	Comments+Criteria
6 (c)	 (i) $\angle ADB = \angle BED = 90^\circ$ (Given) $AB \parallel DE$ (Corresponding angles $\angle ADB = \angle BED = 90^\circ$) $\therefore \angle ABD = \angle BDE = \theta$ (Alternate angles are equal on parallel lines) $\therefore \triangle ABD \sim \triangle BDE$ (Equiangular) (ii) In $\triangle ABD$ $\cos \theta = \frac{BD}{AB}$ $BD = 10 \cos \theta$ as req'd (iii) In $\triangle BDE$ $\cos \theta = \frac{DE}{BD}$ $= \frac{DE}{10 \cos \theta}$ $\therefore DE = 10 \cos^2 \theta$ Similarly: $\triangle DFE \sim \triangle BDE$ as above In $\triangle DEF$ $\cos \theta = \frac{EF}{DE}$ $= \frac{EF}{10 \cos^2 \theta}$ $\therefore EF = 10 \cos^3 \theta$ (iv) $\theta = 30^\circ$ $\therefore EF = 10 \cos^3 30$		1 for progress 1 for answer including statement 1 must show sufficient work 1 for DE 1 for EF

$$= 10 \times \left(\frac{\sqrt{3}}{2}\right)^3 = \frac{30\sqrt{3}}{8} = \frac{15\sqrt{3}}{4} \text{ m}$$

1 correct substitution into (iii) above